Appendix A. Extended Landscape of Existing AI Infrastructure Metric Families

AI infrastructure metrics did not emerge from a single analytical tradition. They were developed in different engineering, computing, and economic communities, often for different operational purposes and under different measurement boundaries. As a result, the current landscape is rich but heterogeneous. The objective of this appendix is not to provide an exhaustive catalogue of all available indicators, but to document the principal metric families that most directly affect capital intensity, operating expenditure, lifecycle cost, risk exposure, productivity, and sustainability-linked financial outcomes in modern AI infrastructure.

This appendix serves two purposes. First, it provides a compact reference map for the major metric families discussed in the main text. Second, it makes explicit why these families, although individually useful, do not by themselves provide a decision-ready system for cross-layer planning, boundary-consistent comparison, or traceable metric synthesis.

A.1. Facility efficiency and resource-overhead metrics

One major family originates in data-center engineering and thermodynamics. These metrics focus on how much non-IT overhead is required to support IT load, especially through power conversion, cooling, water use, and heat rejection. The best-known examples are the xUE family, including power usage effectiveness (PUE), water usage effectiveness (WUE), carbon usage effectiveness (CUE), and energy reuse effectiveness (ERE). Related indicators include cooling power ratio, thermal design margin, and other measures of facility overhead and thermodynamic constraint.

These metrics are economically important because they shape recurring operating cost and infrastructure requirements. A change in PUE, for example, directly changes the amount of purchased electricity required to support a fixed IT load, which in turn affects electricity expenditure, cooling-related OpEx, carbon-valued exposure, and in some cases future cooling CapEx. WUE and related water-intensity measures matter when cooling feasibility is constrained by water availability, water rights, or local environmental regulation.

At the same time, facility metrics are limited by their boundary. They describe physical overhead at the facility level, but they do not explain whether an observed change in energy use or cost is driven by facility design, workload behavior, hardware utilization, grid conditions, or some interaction among them. They are therefore informative but incomplete when the goal is system-level planning rather than isolated facility benchmarking. Table A1 provides a compact cross-family summary of the principal metric traditions, their primary analytical focus, and their decision relevance.

Table A. Economically-Relevant Families of AI Infrastructure Metrics and Their Technical Origins

Metric Family	Originating Discipline	Economic / Financial Relevance	Measurement Boundary	Representative Metrics (Examples)
Facility Efficiency & Resource Overhead Metrics	Data center engineering; ISO/IEC 30134; ASHRAE Datacom	Determines facility OpEx, cooling cost, energy procurement exposure, carbon-embedded cost	Facility physical infra (power, cooling, water, heat rejection)	PUE, WUE, CUE, ERE; Cooling Power Ratio; Thermal Design Margin
Compute Efficiency & Device-Level Performance Metrics	HPC benchmarking; GPU/TPU architecture; accelerator design	Directly shapes compute-per-dollar, job duration, amortization rate of CapEx, marginal cost of compute	Chip, GPU/TPU, node-level compute hardware	FLOPs/W, joules per token, tokens/J, accelerator utilization, thermal throttling indicators
ML System / Workload Execution Metrics	Distributed training systems; MLPerf; model-serving research	Determines productivity cost (developer-time), job completion time (affects OpEx), scheduling efficiency, cost/training-run	Job-level, pipeline-level, distributed runtime	MLPerf throughput; time-to-convergence; inference latency; inference/request; idle energy overhead

Economic & Lifecycle Cost Metrics	Cloud economics; IT finance; energy economics	Captures total lifecycle cost (CapEx/OpEx), depreciation, cost of energy, carbon-adjusted cost, and long-run economic sustainability	Organizational, cluster, region, and workload levels	TCO, LCOE, CapEx-to-OpEx allocation, depreciation curve, marginal cost of compute unit
Reliability, Service Continuity, and Risk Metrics	SRE engineering; reliability theory; operations research	Quantifies risk cost, SLA penalty exposure, redundancy overhead, downtime loss, risk-adjusted TCO	Cluster, fleet, service, and component levels	SLA/SLO compliance; MTBF, MTTR, restart cost, redundancy cost, expected downtime loss
Grid Interaction & Sustainability-Embedded Economics Metrics	Power systems engineering; carbon accounting; energy markets	Determines carbon-adjusted TCO, regional cost arbitrate, demand-response revenue, price volatility risk	Regional grid, power market interface	Marginal Emissions Factor (MEF); LMP/RTM electricity price; carbon price exposure; grid-interaction cost

Notes: This table provides a high-level orientation to the metric families discussed in this section. Detailed critique and structural limitations of these families follow in the paragraphs below. Details of all the metrics are shown in Appendix B, Appendix C (PUE, WUE, CUE, ERE), and Appendix D (TCO series).

A.2. Compute efficiency and device-level performance metrics

A second family comes from high-performance computing, accelerator design, and device benchmarking. These metrics evaluate how efficiently chips, accelerators, and node-level hardware transform electrical power into raw compute capability. Representative examples include FLOPs/W, joules per token, tokens/J, accelerator utilization, temperature-dependent throttling indicators, and related device-level efficiency measures.

Their economic relevance is immediate even when the units are not monetary. Better device efficiency improves compute-per-dollar, affects job duration, influences the amortization rate of expensive accelerator fleets, and changes the marginal cost of useful computation. In modern AI systems, these metrics are also closely tied to memory hierarchy, interconnect behavior, and thermal limits, so "compute efficiency" cannot be understood as a purely chip-level concept detached from system configuration.

However, this family also remains incomplete when used alone. Device-level metrics say little about facility overhead, reliability exposure, carbon boundary choice, or service-level performance. A deployment may look highly efficient at the accelerator level while still producing poor economic outcomes once cooling burden, curtailment exposure, or downstream execution inefficiencies are taken into account.

A.3. ML system and workload-execution metrics

A third family is rooted in distributed systems, ML runtime optimization, benchmarking consortia, and model-serving research. These metrics describe how effectively a software and runtime stack converts installed hardware capability into realized training progress or serving performance. Important examples include throughput, time-to-convergence, latency, inference per request, idle energy overhead, distributed communication overhead, and workload-level efficiency measures such as joules per inference.

MLPerf has become a widely recognized benchmark suite in this area. MLPerf Training emphasizes time-to-convergence under fixed accuracy conditions, while MLPerf Inference emphasizes latency and throughput under specified serving scenarios. These measures matter economically because they shape developer productivity, experimentation speed, model-iteration cost, scheduling efficiency, and serving economics.

Yet workload metrics also sit at a distinct boundary. They capture execution behavior at the job, pipeline, or runtime level, but they do not fully incorporate facility constraints, electricity-market conditions, or lifecycle valuation. They are therefore crucial for performance interpretation but insufficient for broader planning tasks such as siting, procurement, or cross-layer capacity expansion.

A.4. Economic and lifecycle cost metrics

A fourth family comes from cloud economics, IT finance, and energy economics. These metrics are explicitly valuation-oriented. They include total cost of ownership (TCO),

levelized cost concepts such as LCOE, depreciation schedules, CapEx-to-OpEx allocation, marginal cost of compute, and related lifecycle finance measures.

This family is often closest to the language used in executive decision-making because it translates infrastructure decisions into cost, investment burden, and long-run sustainability. TCO frameworks typically incorporate capital investment, electricity procurement, staffing, interconnect, maintenance, refresh cycles, and depreciation. In practice, these models are often the place where engineering quantities are converted into organizationally actionable cost terms.

Their limitation is not lack of relevance but lack of internal endogeneity. In many implementations, engineering efficiency, reliability, and grid conditions appear only as exogenous inputs or scenario assumptions. Complex physical realities are often compressed into a small number of design primitives, such as server density, rack power, accelerator cost, networking topology, and component choices including memory and storage. This compression is useful for financial modeling, but it can obscure the propagation of lower-layer engineering constraints into higher-layer economic outcomes.

A.5. Reliability, service continuity, and operational risk metrics

A fifth family comes from reliability engineering, site reliability engineering, and operations research. These metrics focus on continuity, failure behavior, and service assurance. Common examples include service-level indicators and objectives (SLIs/SLOs), service-level agreements (SLAs), mean time between failures (MTBF), mean time to repair (MTTR), restart cost, redundancy cost, expected downtime loss, and other risk-oriented service metrics.

Their economic significance is substantial. Reliability failures propagate into throughput loss, penalty exposure, redundancy overhead, wasted work, delayed completion, and ultimately risk-adjusted lifecycle cost. In service-facing settings, they also determine whether compute capability translates into usable service delivery under real operating conditions.

Still, these metrics are often measured at different boundaries. Some belong to the runtime and execution layer, where failures manifest as retries, checkpoint overhead, or degraded progress. Others belong to the service layer, where the relevant object is delivered availability, user-facing latency, or contractual compliance. Without a disciplined structure, these reliability measures can be mixed in ways that obscure boundary differences or double count the same failure process at multiple levels.

A.6. Grid interaction and sustainability-embedded economics metrics

A sixth family originates in power systems engineering, carbon accounting, and electricity-market analysis. These metrics capture the external operating environment in which AI infrastructure must function. Representative examples include temporal carbon intensity, marginal emissions factors, renewable availability, demand-response participation value, locational electricity prices, real-time market price exposure, carbon price exposure, and grid-interaction cost.

These quantities are increasingly central in large-load planning because they shape electricity expenditure, carbon-adjusted valuation, demand-response revenue potential, compliance exposure, and the feasible operating envelope of energy-intensive AI clusters. In high-growth regions, grid-facing metrics can determine whether a system is merely expensive, structurally constrained, or exposed to curtailment and reliability risk.

Their limitation is again one of scope. Grid metrics characterize the external environment, but they do not specify how facility design, hardware choices, workload execution, and service commitments mediate the effect of that environment on realized compute delivery or lifecycle value. As a result, they must be linked to downstream layers before they can support planning decisions in a coherent way.

A.7. Why the existing metric families remain insufficient

The problem is not that the existing metric families are individually invalid. The problem is that they were developed under different semantic assumptions, different units, and different operational boundaries. When planning questions involve multiple interacting constraints, this heterogeneity creates recurring failure modes.

First, incompatibility weakens attribution and traceability. A facility metric such as PUE cannot by itself reveal whether an observed cost increase is due to workload behavior, hardware utilization, thermal constraints, or time-varying grid conditions. Conversely, a workload or device metric cannot reveal facility overhead, carbon-cost exposure, or curtailment-related service loss.

Second, overlap and boundary ambiguity create room for both double counting and silent omission. Concepts such as efficiency, throughput, utilization, and reliability may appear at multiple layers but with different denominators and different measurement objects. Without explicit boundary control, organizations can combine metrics that appear related while actually measuring different phenomena.

Third, subsystem-optimal KPI selection can distort system-level decision-making. A locally attractive metric can be improved by shifting burden to another layer. For example, a compute-side efficiency improvement may coincide with higher facility overhead, greater carbon exposure, or larger reliability risk. In the absence of a formal mechanism for validating completeness, non-redundancy, and decision consistency, metric systems become easy to game and difficult to interpret.

Table 2. Representative Metrics in Major AI Infrastructure Metric Families with Various Characteristics

Metric Family	Representative Metric	Units	Temporal Scope	Spatial Scope	Economic / Semantic Focus
Grid & Sustainability	Grid carbon intensity (CI); MEF	gCO_2/kWh	hourly → sub-hourly	regional grid / power market	Carbon-adjusted cost of compute, electricity price, DR participation value
Facility Efficiency	PUE (Power Usage Effectiveness)	ratio (kWh/kWh)	hourly → monthly	datacenter / facility	Cooling OpEx, energy overhead cost, carbon-embedded cost; thermodynamic inefficiency pathways
Compute Efficiency	FLOPs/W or Tokens/J	FLOPs per watt; joules per token	microsecond → second	chip / GPU / TPU / rack	Compute-per-dollar, thermal throttling impacts on OpEx, accelerator amortization efficiency
Workload Efficiency	Joules per inference; Time-to-convergence	J/op; time to accuracy	millisecond → job / batch	job / pipeline / distributed cluster	Productivity cost, pipeline congestion, cost-per-inference, scheduling elasticity
Economic Metrics	TCO; LCOE; marginal cost of compute	USD/year; USD/op	annual → multi-year	cluster / region / org	Lifecycle finance (CapEx→OpEx), depreciation, price exposure, long-run cost sustainability
Reliability & Operations	SLA availability; MTBF/MTTR	% availability; hours	event-driven; daily	fleet / service / component	Downtime loss, SLA penalty risk, redundancy overhead, risk-adjusted TCO

Notes: This table reports representative, not exhaustive, metrics within each family. Individual metrics may differ substantially in units, temporal granularity, and measurement boundaries. The table is intended to illustrate heterogeneity across families rather than internal completeness.

A.8. Implication for the main framework

Taken together, these limitations explain why the existing landscape does not naturally support holistic economic evaluation or multi-objective planning in AI infrastructure. The literature contains many useful metrics, but it rarely provides a common analytical structure for placing them at the correct layer, aligning their semantics, documenting their boundaries, and composing them into interpretable integrated quantities without hidden double counting.

This is the gap addressed by the main framework in the paper. The unified 6×3 taxonomy provides a common coordinate system across layers and semantic domains, while the procedural decision workflow and Metric Propagation Graph provide a disciplined way to select, connect, and synthesize metrics under explicit boundary rules. The role of this appendix is therefore supportive rather than central: it documents the fragmented landscape in fuller form, while the main text focuses on the framework that makes that landscape decision-usable.

Appendix B. Comprehensive Metrics Dictionary for AI Infrastructure Analysis

This appendix provides a unified and systematic dictionary of the metrics used throughout the paper's six-layer, cross-domain AI infrastructure framework. Because the full stack of AI systems spans physical grid interactions, facility-level thermodynamics, computing hardware, networking fabrics, machine-learning execution, service-level reliability, and economic performance, the terminology used in industry and academic literature is highly heterogeneous. Metrics are often defined inconsistently across organizations, including hyperscalers, chip vendors, ML benchmarking groups, and power-systems agencies, creating ambiguity in cross-layer economic or efficiency analysis.

To resolve this fragmentation, Appendix A consolidates all metrics referenced across the 18 analytical cells (6 Layers × 3 Domains) into a single, coherent reference. The appendix includes all metrics.

Table B. Unified Metrics Dictionary Across All Layers and Domains of AI Infrastructure

	Metrics	Definition	Interpretation
Grid & Sustainability Layer - Domain 1: Physical Efficiency	Marginal Emissions Factor (MEF)	$\mathrm{MEF} = \frac{\Delta \mathrm{Emissions}}{\Delta \mathrm{Load}}$	The marginal emissions factor measures the change in grid-level CO_2 emissions from a marginal increase or decrease in electrical load. It reflects the carbon intensity of the marginal generator.
	Time-Varying Grid Carbon Intensity	$\mathrm{CI}(t) = \frac{E_{CO_2}(t)}{P_{grid}(t)}$	Carbon intensity represents the instantaneous emission rate of the grid at time t, generally expressed in gCO_2/kWh. It captures temporal fluctuations in the grid's generation mix.
	Renewable Availability	$\mathrm{RA}(t) = \frac{P_{renewable}(t)}{P_{total}(t)}$	Share of electricity generated from renewable sources (solar, wind, hydro, etc.) at time t. Indicates opportunities for renewable-aligned workload shifting.
Grid & Sustainability Layer - Domain 2: Compute & Workload Efficiency	Grid-Constrained Workload Shiftability	Key metric: Shiftable Workload Fraction (SWF) $\mathrm{SWF} = \frac{W_{shiftable}}{W_{total}}$	Fraction of workload that can be temporally shifted to align with low-carbon or low-cost periods. Measures the workload's temporal flexibility under grid constraints.
	Carbon-Aware Scheduling Limits	Key metric: Emissions-Reduction Potential (ERP) $\mathrm{ERP} = \frac{E_{baseline} - E_{CAS}}{E_{baseline}}$	The maximum proportion of emissions that can be reduced by shifting execution into lower-carbon intervals, assuming perfect knowledge of carbon intensity signals.
	Workload Flexibility Under Grid Conditions	$\mathrm{WF}(t) = \frac{W_{\text{executed in flexible window}}(t)}{W_{total}(t)}$	Represents the share of active workload that responds to real-time grid conditions (curtailment, congestion, or carbon signals). Useful for carbon-aware or demand-response compute.
Grid & Sustainability Layer - Domain 3: Lifecycle Economics & Reliability Risk	Time-Varying Electricity Price (LMP / RTO Pricing)	$\mathrm{Price}(t) = \lambda_{LMP}(t)$	The locational marginal price reflects the marginal cost of serving the next unit of electricity at time t, incorporating energy, congestion, and loss components.
	Carbon Pricing Exposure	$\mathrm{CPE}(t) = \mathrm{CI}(t) \times \mathrm{CarbonPrice}(t)$	Effective monetary exposure to carbon cost at time t. Used when carbon pricing, carbon taxes, or cap-and-trade systems apply.
	Carbon Cost per Token	$C_{carbon,token} = \frac{gCO_2}{\text{token}} \cdot \text{carbon price}$	Monetary cost of carbon emissions attributable to generating one unit of AI output (one token), integrating facility efficiency, grid carbon intensity, and carbon pricing. It reflects how sustainability constraints propagate into model inference cost.
	Location-Based Energy Arbitrage Potential	$\mathrm{AP} = \max_{i,j}(\mathrm{Price}_j)$	Measures the economic advantage of running compute in lower-price regions and shifting away from higher-price regions. Relevant in multi-region data center portfolios.
	Grid Interaction Cost	$C_{interaction} = C_{energy} + C_{demand} + C_{carbon} + C_{DR\ penalties}$	Composite cost of interacting with the grid, including energy, demand, carbon charges, and demand-response penalties. Captures full economic exposure to grid-side fluctuations.
Facility Layer - Domain 1: Physical Performance & Efficiency	GPUs per Rack (GPR) Formula	$\mathrm{GPR} = \frac{N_{GPU}}{N_{racks}}$	The number of GPUs that can be installed in a single rack, constrained by power, cooling, and network topology. A fundamental limit for cluster design and TCO.
	Rack Power Density (RPD)	$\mathrm{RPD} = \frac{P_{rack}}{\text{rack footprint}}$	Measures the maximum deliverable power envelope for a rack and is the primary physical constraint on deploying GPU-dense clusters. High RPD is required for modern AI clusters (30-120 kW/rack).
	Carbon Price Exposure	$\mathrm{CPE} = \mathrm{CUE} \times \mathrm{CarbonPrice}$	Monetary exposure to carbon pricing mechanisms based on emissions-related energy use.
	Cooling System COP (Coefficient of Performance)	$\mathrm{COP} = \frac{Q_{cooling}}{P_{input}}$	Thermodynamic efficiency of cooling systems; higher COP indicates more efficient cooling.
	Cooling Power Fraction	$\mathrm{CPF} = \frac{P_{cooling}}{P_{facility}}$	Fraction of total facility power dedicated to cooling operations.
	Thermal Design Margin (Cooling Headroom)	$\mathrm{TDM} = T_{throttle} - T_{supply}$	Thermal buffer between cooling supply temperature and throttling threshold.
Facility Layer - Domain 2: Compute & Workload Efficiency	Latency Penalty During Facility Events	$\mathrm{LPF} = \frac{L_{event} - L_{nominal}}{L_{nominal}}$	Relative increase in latency due to cooling or power constraints.
	Throughput Degradation Factor (TDF)	$\mathrm{TDF} = \frac{T_{nominal} - T_{event}}{T_{nominal}}$	Fractional throughput loss caused by facility congestion or curtailment.
	Thermal-Throttling Duty Cycle (TTDC)	$\mathrm{TTDC} = \frac{t_{throttled}}{t_{total}}$	Portion of time hardware operates in throttled state due to thermal constraints.
	Workload Curtailment Ratio (WCR)	$\mathrm{WCR} = \frac{W_{curtailed}}{W_{scheduled}}$	Workload volume curtailed due to facility-level constraints.
	Facility-to-Compute Thermal Coupling Coefficient	$\kappa_{FC} = \frac{\partial T_{chip}}{\partial T_{supply}}$	Sensitivity of chip temperature to changes in cooling supply temperature.
Facility Layer - Domain 3: Lifecycle Economics & Reliability Risk	Energy Opex Share	$\mathrm{EOS} = \frac{C_{energy}}{C_{\text{total facility opex}}}$	Share of facility operating cost driven by electricity consumption.
	Cooling Opex Share	$\mathrm{COS} = \frac{C_{cooling}}{C_{\text{total facility opex}}}$	Share of opex attributable to cooling infrastructure and processes.
	Demand-Response Revenue Potential	$\mathrm{DRP} = \sum_t (P_{incentive}(t) \times \Delta L(t))$	Revenue achievable through participation in grid demand-response events.
	Carbon Cost (from CUE × Carbon Price)	$C_{carbon} = \mathrm{CUE} \times \mathrm{CarbonPrice}$	Carbon cost incurred due to emissions-related energy usage.
	Reliability Downtime Cost	$C_{downtime} = \lambda_{failure} \times L_{downtime}$	Expected annual cost of downtime from facility system failures.
Compute Hardware Layer - Domain 1: Physical Performance & Efficiency	Device Power Draw	$P_{device} = V \times I$	Instantaneous electrical power consumption of a compute device; key determinant of thermal load and energy requirements.
	BM Bandwidth per Watt (HBM-BPW)	$\mathrm{HBM\text{-}BPW} = \frac{BW_{HBM}}{P_{memory}}$	Measures the energy efficiency of on-package high-bandwidth memory (HBM). Critical for LLM training workloads where memory bandwidth, not FLOPs, is often the bottleneck.
	Temperature-Performance Coupling	$\gamma_{TP} = \frac{\partial \mathrm{Perf}}{\partial T_{chip}}$	Sensitivity of device performance to chip temperature; captures thermal throttling behavior.
	Thermal Headroom	$\mathrm{TH} = T_{throttle} - T_{chip}$	Temperature margin before throttling begins; indicates thermal safety and cooling adequacy.
	Voltage-Frequency Scaling (VFS) Efficiency	$\eta_{VFS}(f) = \frac{\mathrm{Perf}(f)}{P(f)}$	Energy efficiency of compute under different clock frequencies and voltages; used in power-capping and DVFS policies.
	Idle Power Fraction	$\mathrm{IPF} = \frac{P_{idle}}{P_{TDP}}$	Fraction of TDP consumed during idle state; reflects static inefficiency of hardware when underutilized.
Compute Hardware Layer - Domain 2:	FLOPs per Watt (FLOPs/W)	$\frac{\mathrm{FLOPs}}{P}$	Standard compute-efficiency metric representing computational throughput per watt of power.

Compute & Workload Efficiency	Joules per Operation	$E_{op} = \frac{E_{total}}{N_{operations}}$	Energy required per operation; reciprocal of FLOPs/W.
	Accelerator Utilization	$U = \frac{t_{active}}{t_{total}}$	Fraction of time an accelerator is actively executing workloads; low utilization indicates scheduling or pipeline bottlenecks.
	Tokens per Joule (LLM Efficiency)	$\frac{\text{Tokens}}{E}$	Measures energy efficiency of LLM inference, increasingly used as a standard efficiency metric for transformer models.
	Memory Bandwidth Utilization	$MBU = \frac{B_{used}}{B_{peak}}$	Fraction of available memory bandwidth consumed; captures memory-bound bottlenecks.
Compute Hardware Layer - Domain 3: Lifecycle Economics & Reliability Risk	Hardware TCO Contribution	$TCO_{HW} = C_{capex} + \sum_{t} \frac{C_{opex}(t)}{(r)^t}$	Portion of total cost of ownership attributable to compute hardware (purchase, maintenance, energy, depreciation).
	Memory Power Share (MPS)	$MPS = \frac{P_{memory}}{P_{system}}$	Fraction of total system power consumed by DRAM/HBM. Often 25-40% in modern AI servers (per "AI Tax" paper). Key for cooling, provisioning, and TCO-per-token.
	Depreciation Curve	$V(t) = V_0(1-d)^t$	Hardware value after $t$periods under geometric depreciation rate d.
	Cost per Accelerator-Hour	$C_{acc-hour} = \frac{C_{capex}/L + C_{opex}}{H_{annual}}$	Effective hourly cost for running one accelerator, inclusive of amortization and operating expenses.
	Failure-Driven Replacement Cost	$C_{rep} = \lambda_{fail} \times C_{unit}$	Expected annual replacement cost for failed compute devices; key in reliability-aware economic modeling.
	Hardware Lifetime Efficiency	$HLE = \frac{\text{Useful Compute Output}}{\text{Lifetime Cost}}$	Overall efficiency of hardware over its usable life, defined as compute delivered per dollar spent.
Networking & Interconnect Layer - Domain 1: Physical Performance & Efficiency	Network Power Overhead	$NPO = P_{network} = P_{switches} + P_{NICs} + P_{optics}$	Total power consumed by network infrastructure (switches, NICs, optics). Indicates physical overhead of interconnect relative to compute.
	Network Diameter / Hops	$D = \text{max shortest} - \text{path length between nodes}$	Network Diameter is the maximum shortest path (in hops) between any two accelerators in the cluster topology. Fatal to all-reduce, MoE routing, parameter sharding time lag. LLM inference + MoE routing = latency highly sensitive to network diameter
	Switch Load Power Ratio	$SLPR = \frac{P_{switch}}{P_{rack}}$	Fraction of rack-level power used by switching hardware; reflects network intensity of a cluster.
	Rack/Pod Thermal Load from Networking	$TL_{net} = \frac{P_{network}}{P_{total}}$	Share of thermal load contributed by networking equipment; influences cooling planning and hot-aisle design.
	Interconnect Energy per Bit	$E_{bit} = \frac{E_{network}}{B_{transmitted}}$	Energy required to transmit one bit over the fabric; key for large-scale distributed training.
Networking & Interconnect Layer - Domain 2: Compute & Workload	Pipeline Efficiency (PE)	$PE = \frac{T_{ideal}}{T_{actual}}$	Measures how efficiently a distributed pipeline (training or inference) achieves ideal end-to-end throughput. Captures pipeline bubbles and imbalance.
	Bisection Bandwidth（BB/sec）	$BB/sec = \frac{IBB}{2}(\text{cluster evenly partitioned}) = \min_{\text{cluster bisection}} \sum_{\ell} B\,W_{\ell}$	Classical HPC bandwidth metric used to characterize communication capacity under worst-case workload partitioning. High BB is essential for large-model all-reduce and for maintaining low synchronization overhead in transformer training.
	Interconnect Bisection Bandwidth (IBB)	$IBB = \min_{\text{cut}} \sum_{\text{links crossing cut}} \text{bandwidth}$	Measures the minimum bandwidth available across any cluster-wide network bisection and is the dominant determinant of distributed training scalability. Essential for superpods design.
	Bandwidth per Watt	$BPW = \frac{B}{P_{network}}$	Data-movement throughput per watt of networking power. Higher values indicate more energy-efficient interconnect hardware.
	All-Reduce Latency (ARL)	$L_{collective} == f(\text{topology, message size, congestion})$	Latency of collective communication (AllReduce, Broadcast, etc.), central to distributed deep learning performance.
	Congestion Penalty (CP)	$CP = \frac{L_{p99} - L_{p50}}{L_{p50}}$	Quantifies tail-latency degradation due to network congestion; reflects imbalance and queueing delays under heavy load.
	Topology-Induced Throughput Loss	$TTL = 1 - \frac{T_{observed}}{T_{full-bisection}}$	Fraction of throughput lost because real-world topology deviates from ideal full-bisection bandwidth (e.g., oversubscription).
	Effective Bandwidth Utilization	$EBU = \frac{B_{useful}}{B_{peak}}$	Efficiency of available network bandwidth actually used for productive compute-related communication.
Networking & Interconnect Layer - Domain 3: Lifecycle Economics & Reliability Risk	Interconnect CapEx Contribution	$C_{net,capex} = C_{switches} + C_{optics} + C_{NICs}$	Capital cost associated with networking hardware as part of a cluster or data center build-out.
	Network Scaling Cost	$C_{scale} == f(N, \text{fabric type, oversubscription ratio})$	Total cost of scaling the network as cluster size Ngrows; reflects topology, bisection needs, and redundancy requirements.
	Topology-Driven Efficiency Loss	$TDL = \frac{T_{ideal} - T_{topology}}{T_{ideal}}$	Portion of throughput lost due to non-ideal network topology design (e.g., fat-tree oversubscription).
	Network Failure Rate	$\lambda_{net} = \frac{\text{Number of failures}}{\text{Time}}$	Frequency of network-component failures (optics, NICs, switches); affects reliability planning and redundancy needs.
	Network Reliability Cost	$C_{reliability} = \lambda_{net} \times L_{event}$	Expected economic cost associated with network outages or degraded service.
ML Execution Layer - Domain 1: Physical Performance & Efficiency	Thermal-Induced Failure Rate	$\lambda_{thermal} = f(T_{chip}, t_{exposure})$	Failure rate caused by prolonged exposure to elevated chip temperatures. Often modeled using Arrhenius acceleration factors.
	Power-Quality-Induced Failure Rate	$\lambda_{PQ} = f(\text{voltage sag, harmonics, transients})$	Failure frequency arising from unstable or degraded power quality, including brownouts and voltage spikes.
	Compiler-Induced Energy Overhead	$CEO = \frac{E_{compiled} - E_{ideal}}{E_{ideal}}$	Additional energy consumption caused by suboptimal compiler or kernel scheduling decisions.
	Idle Energy from Inefficient Scheduling	$E_{idle} = P_{idle} \times t_{idle}$	Total idle energy burned due to scheduling stalls, synchronization delays, or uneven pipeline workloads.
ML Execution Layer - Domain 2: Compute & Workload Efficiency	MLPerf Time-to-Train (TTT)	$TTT = t_{converged} - t_{start}$	Time required for a model to reach a defined target accuracy. Primary metric for MLPerf Training benchmark.
	MLPerf Throughput (QPS/TPS)	$TPS = \frac{\text{Total Queries}}{t_{window}}$	Inference throughput under MLPerf constraints; widely used industry benchmark.
	Data Pipeline Stalling Rate (DPSR)	$DPSR = \frac{\text{Stalled Steps}}{\text{Total Steps}}$	Fraction of training steps where GPUs stall waiting for input data (decode, augment, batch assemble, prefetch).
	Job Packing Efficiency/Fragmentation Loss	$\text{Fragmentation Loss} = 1 - JPE$	Utilization loss not caused by compute or data bottlenecks, but caused by orchestration geometry
	Effective Model Parallel Efficiency (MPE)	$MPE = \frac{\text{Achieved Throughput}}{\text{Ideal Parallel Throughput}}$	It captures communication overhead, pipeline bubbles, imbalance among stages, and tensor-parallel sharding cost.
	Tokens per Second (LLM Serving Throughput)	$TPS_{LLM} = \frac{\text{Tokens}}{t}$	Standard measure of LLM decoding throughput.

	Storage I/O Throughput per Training Step	$IO_{per\ step} = \frac{V_{IO}}{N_{steps}}$	Measures dataset read bandwidth and I/O system ability to feed GPUs timely. GPU idle waiting for next batch.
	Pipeline Parallelism Efficiency	$PPE = \frac{T_{ideal}}{T_{pipeline}}$	Captures pipeline bubbles and per-stage imbalance in multi-stage distributed execution.
	Energy per Checkpoint (EPC)	$EPC = \frac{E_{checkpoint}}{N_{checkpoints}}$	Energy consumed per model checkpoint during training.
ML Execution Layer - Domain 3: Lifecycle Economics & Reliability Risk	Cost per Training (TPT)	$C_{step} = \frac{C_{total}}{N_{steps}}$	Total cost (compute, energy, depreciation) amortized per training step.
	Cost per Token (Inference)	$C_{token} = \frac{C_{inference}}{N_{tokens}}$ or $C_{token} = \frac{P_{GPU} \cdot t_{lat}}{tokens}$	Monetary cost of generating one token during inference.
	Marginal Serving Cost (MSC)	$MSC = \frac{\partial C}{\partial Q}$	Additional cost of serving one more unit of workload (e.g., per 1k tokens or per request).
	Autoscaling Overhead (ASO)	$ASO = \frac{C_{autoscale} - C_{ideal}}{C_{ideal}}$	Cost penalty due to autoscaling inefficiencies such as cold starts or idle buffer capacity.
	Expected Loss from Training Instability	$EL_{inst} = P_{diverge} \times C_{restart}$	Expected loss from training runs that diverge and must be restarted.
	Annual Loss Expectancy (ALE) from Failures	$ALE = \lambda_{failure} \times L_{event}$	Expected annual economic loss resulting from system failures (hardware, network, software).
Service Layer - Domain 1: Physical Performance & Efficiency	Redundancy Energy Overhead	$REO = \frac{P_{redundant}}{P_{total}}$	Fraction of energy spent on redundant components (e.g., N+1, 2N systems) to ensure service reliability.
	Energy from Thermal Penalties (Derating Overhead)	$DEO = \frac{E_{derated} - E_{optimal}}{E_{optimal}}$	Extra energy incurred when compute systems operate below optimal efficiency due to high temperatures or thermal derating.
	Service Energy Inefficiency Factor	$SEIF = \frac{E_{service}}{E_{compute}}$	Overhead energy consumption required for achieving a given level of service quality (e.g., replication, retries).
	Thermal Stability Index	$TSI = \frac{T_{max} - T_{service}}{T_{max} - T_{nominal}}$	Normalized indication of thermal stability of service nodes relative to safe operating bounds.
Service Layer - Domain 2: Compute & Reliability (SLO/SLA Metrics)	Service Level Indicator (SLI)	$SLI = \frac{\text{Number of successful requests}}{\text{Total requests}}$	Base measurement of service correctness, availability, or latency adherence.
	Availability (A)	$A = \frac{\text{Uptime}}{\text{Uptime} + \text{Downtime}}$	Standard measurement of system availability; often used in "five nines" reliability targets.
	SLO Violation Rate	$\text{ViolationRate} = \frac{\text{Violations}}{\text{Total Requests}}$	Percentage of requests failing to meet defined service-level objectives (latency, throughput, correctness).
	Error Budget Consumption (EBC)	$EBC = \frac{SLO - SLI}{1 - SLO}$	Fraction of allowable error budget consumed; widely used in SRE practice.
	Mean Time Between Failures (MTBF)	$MTBF = \frac{\text{Total Operating Time}}{\text{Number of Failures}}$	Frequency metric describing how often service-impacting failures occur.
	Mean Time to Recovery (MTTR)	$MTTR = \frac{\sum \text{Repair Time}}{\text{Number of Failures}}$	Average time to restore service after an outage.
	QoS Compliance Rate	$QCR = 1 - QDF = 1 - \frac{t_{QoS\ violated}}{t_{total}}$	Fraction of time service meets required QoS/SLO standards.
Service Layer - Domain 3: Lifecycle Economics & Reliability Risk	Total Service Cost (TSC)	$TSC = C_{compute} + C_{network} + C_{energy} + C_{SRE} + C_{failures}$	Full economic cost of providing service reliability and performance, including SRE labor, hardware, network, and failure losses.
	GPU Interconnect Oversubscription Ratio	$OSR = \frac{\text{GPU aggregate BW}}{\text{switch uplink BW}}$	Measures how heavily the GPU-to-switch or GPU-to-GPU network bandwidth is oversubscribed relative to the aggregate bandwidth required by training workloads.
	SLA Penalty Cost	$C_{SLA} = \sum_i (L_i \times P_{penalty})$	Cost incurred when service availability breaches contractual SLAs.
	Cluster Utilization Rate (CUR)	$\mathbf{CUR} = \frac{\text{Active Compute Time}}{\text{Total Provisioned GPU Time}}$	Measures the fraction of total available accelerator capacity actively performing useful computation over time, determining GPU amortization efficiency.
	Token Throughput per Watt (TPW)	$\mathbf{T}PW = \frac{\text{Tokens Processed}}{\text{Energy (W\cdotps)}}$	Energy efficiency metric for inference (and sometimes training). Fits better for LLM inference than FLOP/W. Key for inference scaling, cost-per-token, and carbon footprint.
	Expected Loss from MTBF/MTTR Mismatch (ALE)	$ALE = \lambda_{failure} \times L_{event}$	Expected annual financial loss due to reliability shortfalls (hardware, network, or service components).
	Operational Cost Elasticity	$OCE = \frac{\partial C_{service}}{\partial Q}$	Additional service cost incurred per incremental unit of workload (queries, tokens, users).
	Service Lifetime Value (SLV)	$SLV = \sum_{t=1}^{T} \frac{\text{Revenue}(t) - \text{Cost}(t)}{(r)^t}$	Net present value of operating a service over its lifetime.

Notes: This table consolidates all metrics defined across 6x3 taxonomy into a single, unified reference spanning the full six-layer, three-domain analytical framework. Metrics include physical, thermal, compute-level, networking, ML execution, reliability, and economic indicators used throughout the paper's propagation-graph and cross-layer analysis. Each entry preserves the original terminology used in Table 3 while providing a standardized mathematical formulation and concise interpretation for reproducibility. Metrics belonging to the xUE family (e.g., PUE, WUE, CUE, ERE) and TCO-related measures are omitted here, as they are documented separately in dedicated appendices.

Appendix C. A PUE Literature Review

Power Usage Effectiveness (PUE) remains one of the most important metrics for evaluating data-center energy efficiency, originally introduced by The Green Grid in 2007 [56-60]. Since its

adoption, PUE has been widely used as a high-level indicator of facility overhead, and continues to appear as the dominant metric in recent industrial assessments. For example, the Uptime Institute's global survey reports a nearly flat annual average PUE of about 1.56 for major facilities (2024), reflecting both historical progress and ongoing stagnation across legacy sites [61].

At the same time, a growing literature highlights the limitations of PUE, especially for high-density, AI-intensive deployments. Li et al. (2020) demonstrate that PUE alone cannot support fair cross-site comparison due to sensitivity to climate, IT utilization, and design strategies, and propose a normalized indicator that decomposes these external factors [62]. Earlier critiques, such as Horner and Azevedo (2016), argue that PUE is often "overloaded and overvalued" when used as a proxy for sustainability because it ignores utilization, embodied energy, and workload-level efficiency [63].

These limitations have motivated a shift toward a PUE-family of metrics. The U.S. DOE's Best Practices Guide (2024) explicitly situates PUE within a broader suite of indicators including ERE, WUE, and CUE, and recommends joint monitoring with IT utilization and cooling-system performance [64]. In parallel, broader reviews such as Safari et al. (2025) classify more than twenty-five data-center energy metrics into infrastructure-level, IT-level, and application-level categories, and map their applicability to cloud and AI workloads. Thermal-management surveys such as Xu et al. (2023) likewise emphasize that PUE is useful for comparing cooling topologies but insufficient for capturing interactions between workload characteristics and system-level energy behavior [65].

A second strand of literature uses PUE to quantify the impact of advanced cooling architectures, particularly those capable of pushing PUE toward 1.0 or even lower. Wang et al. (2022) design a directly air-cooled compact looped heat pipe (LHP) module capable of dissipating 550 W per server with minimum thermal resistance of 0.16 °C/W, achieving PUE values as low as 1.02 [51]. Zhou et al. (2022) simulate a liquid-pump-driven hybrid free-cooling and mechanical-cooling system across 20 sub-climate zones and 37 cities, showing annual PUE below 1.40 in most locations except hot-humid regions [58]. Building on these strategies, Zhou et al. (2023) combine an LHP with a thermoelectric generator (TEG) module, achieving PUE = 0.997 at 155 W heat load, demonstrating the technical feasibility of "PUE < 1" under specific operating conditions [59].

AI workloads magnify these issues. The IEA 4E program's Data Centre Energy Use review (2025) estimates that AI accelerators currently account for only 5-15% of global data-center electricity use but may rise to 200-400 TWh by 2030, driven largely by cooling burdens and assumptions about achievable PUE trajectories [66]. These projections warn that relying solely on optimistic PUE improvements may underestimate the energy footprint of upcoming GPU-dense clusters. Meanwhile, industry analyses (2023-2024) report typical PUE ≈ 1.58 for conventional colocation facilities but <1.3 for AI-optimized deployments, underscoring both the performance gap and the uneven adoption across the installed base [61].

To sum up, this body of work, spanning foundational critiques (2016-2020), systematic surveys (2023-2025), advanced cooling demonstrations (2022-2023), and macro-level projections (2024-2025), reveals three consistent insights. First, PUE remains essential but insufficient: indispensable for tracking facility overhead yet blind to workload-level performance and utilization. Second, contemporary best practices rely on tiered, multi-metric frameworks, where PUE/ERE/WUE/CUE form the infrastructure layer, IT utilization and server efficiency the middle layer, and application-level metrics (e.g., energy per inference or per training run) the top layer. Third, although experimental systems can achieve PUE near, or even below, 1.0, the global installed base is dominated by facilities operating around 1.5-1.6, meaning the real-

world energy footprint of AI workloads depends primarily on scaling behavior rather than best-case cooling efficiency.

In this paper, this study adopts this multi-layered perspective by treating PUE and its extensions as infrastructure-level AI metrics and linking them explicitly to workload-level indicators. This allows us to disentangle physical-infrastructure efficiency from algorithmic or hardware efficiency, and to analyze how different AI-infrastructure design choices jointly determine the "PUE × performance-per-watt" frontier.

Table C. Summary of xUE family metrics.

Metric	Variable	Formula	Variables Explanation	Range & Direction	Application Scenarios
Power Usage Effectiveness	PUE	$PUE = \frac{E_{Total\ Facility}}{E_{IT\ Equipment}}$ $E_{Total\ Facility}$ = total energy consumed by the entire data center (kWh). $E_{IT\ Equipment}$ = energy consumed by IT equipment over the same period (kWh).	Measures how efficiently a data center uses energy. Numerator includes IT, cooling, power distribution losses, lighting, etc.	Range: PUE ≥ 1 (theoretical minimum 1, no strict upper bound). Meaning: Lower is better; closer to 1 → higher efficiency.	Global key performance indicator for data center energy efficiency; used in design benchmarking, operations monitoring, and regulatory reporting.
Data Center infrastructure Efficiency	DCiE	$DCiE = \frac{E_{IT}}{E_{Total\ Facility}} = \frac{1}{PUE}$ $E_{Total\ Facility}$ = total energy consumed by the entire data center (kWh). $E_{IT\ Equipment}$ = energy consumed by IT equipment over the same period (kWh).	Expresses the fraction of total facility energy actually delivered to IT equipment. It is the reciprocal of PUE.	Range: 0 < DCiE ≤ 1. Meaning: Higher is better; values closer to 1 indicate a larger share of energy going to IT work.	Used alongside PUE for data center efficiency reporting; sometimes preferred for expressing "percentage of useful energy" to management.
Water Usage Effectiveness	WUE	$WUE = \frac{V_{Annual\ Site\ Water}}{E_{IT\ Equipment}}$ $V_{Annual\ Site\ Water}$ = annual water consumption attributable to the data center (L or m³ per year). $E_{IT\ Equipment}$ = energy consumed by IT equipment over the same period (kWh).	Measures how much water is used per unit of IT energy. Can be reported as L/kWh, m³/MWh, or similar.	Range: WUE ≥ 0. Meaning: Lower is better; smaller values indicate less water used per unit of IT work.	Assessment of water footprint of data centers; comparison of cooling technologies (evaporative vs. air-cooled); environmental and ESG reporting.
Carbon Usage Effectiveness	CUE	$CUE = \frac{M_{CO2}}{E_{IT}}$ or equivalently: $CUE = PUE \times CEF$ Where M_{CO2} = total annual CO_2-equivalent emissions associated with the data center (kg CO_2e/year). E_{IT} = annual IT equipment energy consumption (kWh/year). CEF = carbon emission factor of the electricity supply (kg CO_2e/kWh).	Measures greenhouse gas emissions per unit of IT energy. Using CUE = PUE × CEF explicitly connects facility efficiency with grid carbon intensity.	Range: CUE ≥ 0. Meaning: Lower is better; smaller values indicate lower carbon intensity of operations.	Used for carbon and ESG accounting for data centers; comparing sites with different grid mixes; evaluating decarbonization strategies (renewables, PPAs, etc.).
Energy Reuse Effectiveness	ERE	$ERE = \frac{E_{Total\ Facility} - E_{Reuse}}{E_{IT\ Equipment}}$ or equivalently: $ERE = (1 - ERF) \times PUE$ $E_{Total\ Facility}$ and $E_{IT\ Equipment}$ are the same as before. E_{Reuse} = portion of data center energy that is beneficially reused outside the data center (kWh). $ERF = \frac{E_{reuse}}{E_{Total\ Facility}}$ $(energy\ reuse\ factor)$.	Adjusts PUE to account for beneficial reuse of waste heat. Lower ERE indicates higher overall effective efficiency when reused energy is considered.	Range: 0 ≤ ERE ≤ PUE. Meaning: Lower is better; an ERE less than PUE reflects effective energy reuse.	Evaluation of data centers that export waste heat (district heating, building heating, industrial processes); used when assessing integrated energy systems and circular energy.

Appendix D. Comprehensive Mapping of TCO-Series Metrics, Formulas, and Economic Semantics

Total Cost of Ownership (TCO) has long served as the foundational economic framework for evaluating data-center and compute-infrastructure investments. The earliest formal treatments, most notably Koomey (2007) and Rasmussen (2005), established TCO as a lifecycle accounting model that captures capital expenditure, operational expenditure, energy consumption, cooling requirements, power-distribution overhead, and infrastructure depreciation [67]. These works argue that traditional cost accounting underestimates long-term facility and energy costs and propose a "true TCO" model that aligns financial planning with engineering realities.

As data-center engineering matured, TCO variants emerged to evaluate cost intensity at different levels of physical granularity. Metrics such as TCO/kW, TCO/Rack, and TCO/kWh became widely used in colocation pricing, energy-storage economics, and data-center design, particularly within APC/Schneider Electric white papers and NREL cost-assessment tools [67-69]. These variants highlight how power density, rack utilization, cooling topology, and site electrical configuration reshape the amortized cost structure of large-scale compute facilities.

Broader energy-sector economics contributed additional levelized metrics. The levelized cost of electricity (LCOE), formalized through NREL and EIA methodologies [69,70], frames electricity price as a discounted lifecycle quantity rather than an instantaneous rate. The levelized cost of hydrogen (LCOH), widely used in energy-transition analysis [71,72], introduced standardized techniques for comparing heterogeneous energy carriers. These methods influenced data-center economics by providing a consistent way to evaluate long-run marginal cost under technology turnover, fuel-price volatility, and policy-driven cost shifts.

With the rise of AI and HPC systems, TCO analysis expanded beyond physical infrastructure to computational output. Metrics such as TCO per GPU·hour, TCO per FLOP, TCO per inference, and TLOps (total levelized operations) emerged to quantify the economic efficiency of compute capacity under modern accelerator architectures. While early HPC economics treated compute cost implicitly, contemporary AI workloads, characterized by high GPU density and steep scaling, require explicit accounting of communication overhead, parallelization constraints, and failure-recovery cost, all of which materially affect amortized cost-per-operation.

A major recent development is the Levelized Cost of AI (LCOAI) [31]. LCOAI extends levelized-energy concepts into AI computation by integrating hardware depreciation, accelerator refresh cycles, cluster topology efficiency, failure rates, power-usage effectiveness, and energy-procurement cost into a unified lifecycle metric. It also formalizes cost-per-inference and cost-per-training-run as levelized quantities comparable across hardware generations, cloud platforms, and deployment environments.

Finally, newer conceptual metrics such as TECO (Total Environmental Cost of Ownership) and MTCOF (Marginal TCO of Compute Flow) reflect the growing need to incorporate sustainability, carbon pricing, and marginal-cost modeling into infrastructure economics [69]. Although less standardized, these extensions signal an emerging direction where TCO integrates operational resilience, ESG constraints, and multi-carrier energy interactions.

Table D. Summary of TCO-Series Metrics and Their Economic Interpretation

Metric	Variable	Formula	Variables Explanation	Range & Direction	Application Scenarios
Total Cost of Ownership	TCO	$TCO = C_{CAPEX} + \sum_{t=1}^{T} \frac{C_{OPEX,t}}{(1+r)^t}$ r = discount rate; T = analysis horizon (All notations work the same to all other terms below)	The discounted sum of all capital and operating costs over the lifetime, capturing one-time and recurring costs in a single financial metric.	Range: TCO ≥ 0. Direction: Lower TCO is preferable for a given level of capacity or service.	Overall budgeting for data centers and AI infrastructure; comparing design options; evaluating build vs. lease decisions.
TCO per kW of IT Capacity	TCO/kW	$TCO\ per\ kW = \frac{TCO}{P_{IT,rated}}$ $P_{IT,rated}$ = rated IT load capacity of the facility (kW)	Lifecycle cost per unit of installed IT power capacity. Used by colocation providers and operators to compare design and site options.	Range: TCO/kW ≥ 0. Direction: Lower is better for a given quality-of-service and uptime target.	Benchmarking cost-efficiency of data centers; comparing wholesale colocation offers; evaluating power density strategies.
TCO per kWh of Delivered Energy	TCO/kWh	$TCO\ per\ kWh = \frac{TCO}{E_{total}}$ E_{total} = total energy delivered or consumed over the period (kWh)	Lifecycle cost per unit of energy delivered or processed, esp. for evaluating and comparing long-lived energy or storage assets.	Range: TCO/kWh ≥ 0. Direction: Lower is better; compared to market electricity prices or competing technologies.	Assess economic competitiveness of batteries, fuel cells, backup generators, and other power solutions for data centers.
TCO per Rack	TCO/Rack	$TCO\ per\ Rack = \frac{TCO}{N_{rack}}$ N_{rack} = number of IT racks served by the facility	Lifecycle facility cost allocated per rack. Used to compare the cost of housing IT equipment in different facilities or designs.	Range: TCO/Rack ≥ 0. Direction: Lower is better for a given service level and rack power density.	Planning and pricing colocation services; internal budgeting for IT equipment deployment; comparing legacy vs. modernized facilities.
TCO per GPU-Hour	TCO per GPU·h	$TCO\ per\ GPU \cdot h = \frac{TCO}{H_{GPU}}$ H_{GPU} = total GPU-hours delivered to workloads in that period	Lifecycle cost per GPU-hour of compute delivered. Combines capex and opex with utilization as an effective cost per GPU-hour.	Range: ≥ 0. Direction: Lower is better; provides a direct comparison to cloud GPU-hour pricing.	Compare on-premises GPU clusters vs. cloud GPUs; optimizing cluster utilization; cost benchmarking across vendors.
TCO per FLOPs	TCO/FLOPs	$TCO\ per\ FLOPS = \frac{TCO}{F_{total}}$ F_{total} = total floating point operations (FLOPs) delivered over the analysis period	Lifecycle cost per unit of compute (FLOPs). Can be defined using aggregated FLOPs across training and inference workloads.	Range: ≥ 0. Direction: Lower is better; allows comparing different architectures and accelerators in a normalized way.	Compare GPUs, TPUs, and custom accelerators; evaluating architectural trade-offs for large-scale AI workloads.
TCO per Inference	TCO/inference	$TCO\ per\ Inference = \frac{TCO}{N_{inf}}$ N_{inf}= total number of valid inferences (e.g., requests, queries, responses) served in t	Lifecycle cost per unit of AI service delivered. Can be normalized per 1,000 inferences.	Range: ≥ 0. Direction: Lower is better; directly links infrastructure	Price AI APIs and services; evaluating deployment options (vendor APIs vs. self-hosted models); optimizing autoscaling and provisioning.

				economics to business KPIs like cost per API call.	
True Total Cost of Ownership	True TCO	$True\ TCO = \Sigma_{\{t=0\}}^{\{T\}} \frac{C_{f,t} + C_{i,t} + C_{e,t} + C_{s,t} + C_{o,t}}{(1+r)^t}$ $C_{f,t}$ = facility-related costs (construction, space, cooling, power) $C_{i,t}$ = IT hardware and software costs $C_{e,t}$ = energy costs $C_{s,t}$ = personnel costs $C_{o,t}$= other recurring or indirect costs	Extends basic TCO by explicitly including all relevant facility, IT, and indirect costs. Often implemented via spreadsheet models for scenario analysis.	Range: ≥ 0. Direction: Lower True TCO is better for a given level of service; the metric is mainly used for comparison.	Strategic planning for data center investments; compare high-efficiency vs. conventional designs; support executive decision-making.
Total Cost of Energy	TCOE	$TCOE = \Sigma_{\{t=1\}}^{\{T\}} \frac{C_{energy,t}}{(1+r)^t}$ $C_{energy,t}$ = energy-related costs in year t (electricity, demand charges, fuel)	Isolates the energy component of lifecycle cost. Extended to include energy-related infra (on-site generation or storage).	Range: ≥ 0. Direction: Lower is better; used to assess and compare energy strategies.	Compare energy sourcing options; assess the impact of efficiency measures and PUE improvements; evaluating on-site renewables and storage.
Total Environmental Cost of Ownership	TECO	$TECO = TCO + C_{carbon} + C_{water} + C_{other_{env}}$ C_{carbon} = cost of greenhouse gas emissions (e.g. carbon price) C_{water} = cost of water usage and scarcity impacts C_{other_env} = cost of other externalities (land, waste etc.)	Adds monetized environmental externalities to standard TCO, producing an augmented lifecycle cost with ESG priorities.	Range: ≥ 0. Direction: Lower is better; encourages designs that minimize both financial and environmental impacts.	ESG-focused infrastructure planning; evaluating low-carbon and low-water designs; internal carbon pricing analyses.
Levelized Cost of Energy	LCOE	$LCOE = \Sigma_{\{t=0\}}^{\{T\}} \frac{I_t + M_t + F_t}{(1+r)^t} \div \Sigma_{\{t=0\}}^{\{T\}} \frac{E_t}{(1+r)^t}$ I_t= investment (CAPEX) in year t M_t = operations and maintenance (O&M) costs in year t F_t = fuel costs in year t (if applicable) E_t = electricity in year t (MWh)	Discounted total cost of building and operating a generation asset, divided by discounted total electricity generation (reported in $/MWh).	Range: LCOE ≥ 0. Direction: Lower LCOE indicates more cost-competitive generation, holding other factors equal.	Compare different power generation technologies; evaluating renewable energy projects; policy analysis and planning.
Levelized Cost of Hydrogen	LCOH	$LCOH = \Sigma_{\{t=0\}}^{\{T\}} \frac{C_t}{(1+r)^t} \div \Sigma_{\{t=0\}}^{\{T\}} \frac{H_t}{(1+r)^t}$ C_t = total annual costs in year t (CAPEX recovery, OPEX, energy input costs) H_t = hydrogen produced in year t (kg or Nm^3)	Discounted total cost of producing hydrogen divided by discounted total hydrogen output. Reported in $/kg H2.	Range: LCOH ≥ 0. Direction: Lower LCOH indicates more cost-competitive hydrogen production.	Evaluate and compare green and blue hydrogen projects; (electrolysis, SMR with CCS); investment analysis.
Levelized Cost of Artificial Intelligence	LCOAI	$LCOAI = \frac{CAPEX_{amortized} + \Sigma_{\{t=1\}}^{\{T\}} \frac{OPEX_t}{(1+r)^t}}{N_{valid}}$ $CAPEX_{amortized}$ = amortized capital expenditures for AI systems (hardware, initial training, deployment) $OPEX_t$ = operational expenditures in year t (compute, storage, networking, maintenance, labor) N_{valid}= total number of valid inferences (or other chosen AI output unit) over the analysis period	Lifecycle cost per unit of productive AI output. Directly analogous to LCOE and LCOH, but with "valid inferences" or similar AI outputs as the denominator.	Range: LCOAI ≥ 0. Direction: Lower LCOAI indicates more cost-efficient AI deployment for a given output quality.	Compare vendor APIs vs. self-hosted models; evaluating different deployment architectures; guiding procurement and infrastructure planning for AI workloads.
Total Lifecycle Operations per Cost (TLOps / TLOP)	TLOps (or TLOP)	$TLOps = \frac{Q_{ops}}{TCO}$ Q_{ops} = total quantity of operations delivered (e.g., FLOPs, inferences, tokens) Equivalently, a cost-based form is: $Cost\ per\ operation = TCO\ /\ Q_ops$	Operations delivered per unit of lifecycle cost, or conversely cost per unit of operation. Can be tailored to FLOPs, inferences.	Range: TLOps ≥ 0. Direction: Higher TLOps (more operations per dollar) is better; equivalently, lower cost per operation is better.	Compare different AI clusters or architectures on a cost-normalized basis; assess the impact of utilization and efficiency improvements.
Marginal TCO of FLOPs	MTCOF	$MTCOF = \frac{\partial TCO}{\partial F_{total}}$ F_{total}= total FLOPs delivered In discrete form, it is often approximated as: $MTCOF \approx \frac{\Delta TCO}{\Delta F_{total}}$	Marginal cost of delivering additional compute, used for evaluating scaling decisions (adding more GPUs or moving to a different architecture).	Range: MTCOF ≥ 0 for normal systems. Direction: Lower marginal cost per additional FLOP is better; helps identify cost-effective scaling.	Decide when to expand clusters; compare incremental upgrades vs. full system replacements; assess the value of efficiency improvements.

Notes: All TCO-series metrics in this table, including TCO, TCO/kW, TCO/kWh, TCO/Rack, TCO per GPU·h, TCO per FLOP, TCO per inference, True TCO, TCOE, TECO, LCOE, LCOH, LCOAI, TLOps, and MTCOF, are derived from established cost-modeling frameworks in data-center economics [67-72]. These metrics build upon lifecycle financial analysis, levelized energy economics, and recent AI/HPC cost-per-compute formulations. They collectively provide a unified basis for evaluating the long-run economic efficiency of AI and data-center infrastructure.

Appendix E. Formal Justification of the Three Meta-Metric Domains

The three domains introduced in this work, Physical Facility Efficiency, Compute & Workload Efficiency, and Economic & Reliability Efficiency, qualify as meta-metric domains because they each correspond to a fundamentally distinct class of phenomena governed by independent physical laws, computational models, and economic principles. The domains reflect intrinsic separations in the underlying physical, computational, and economic processes, rather than an imposed or convenience-based taxonomy. Metrics such as PUE, WUE, CUE, or cooling-system coefficients quantify thermodynamic and mechanical conversion processes: the transformation

of grid energy, water, and carbon into usable IT power under constraints defined by heat transfer, airflow mechanics, and facility-scale energy balances. These processes are governed by physical laws independent of the algorithms, computational workloads (e.g., online temporal models [73], or cost models that operate on top of them. In contrast, compute and workload metrics describe the execution behavior of software and models, FLOPs, tokens, batches, latency distributions, and scaling dynamics, reflecting the logic of computational graphs, parallelism strategies, and hardware acceleration. Neither the thermodynamic state of a cooling system nor the airflow efficiency of a data hall can determine model throughput, and conversely, no purely computational measure can capture physical loss mechanisms. Economic and reliability metrics, meanwhile, quantify lifecycle cost, depreciation, risk, service continuity, and financial exposure. These outcomes depend on institutional cost models, market structures, and risk formulations, rather than physical or computational quantities. No domain can be reduced to another without introducing an external modeling layer, and thus they remain conceptually independent.

Second, the three domains differ in their dimensional systems, making them semantically and mathematically incompatible. Physical facility metrics express their behavior in units such as kWh/kWh, L/kWh, gCO_2/kWh, °C/W, or COP, ratios and coefficients that quantify energy and resource flows. Compute and workload metrics operate in a fundamentally different numerical universe: FLOPs, Joules per operation, inference latency in milliseconds, or throughput in sequences per second. These correspond to algorithmic progress, computational effort, and performance semantics rather than energy or resource conversion. Economic and reliability metrics employ yet another dimensional system: USD/year, levelized cost per inference, depreciation rates, or failure probabilities. The three domains operate on incommensurable dimensional systems, physical ratios, computational units, and monetary or probabilistic quantities, precluding any direct mapping without external modeling assumptions.

Third, the domain decomposition is both complete and minimal. Every known AI infrastructure metric can be mapped unambiguously to one, and only one, of the three domains. Metrics describing energy, thermal behavior, water use, and mechanical overhead belong to the physical facility domain; metrics describing model execution, accelerator performance, and workload throughput belong to the compute and workload domain; metrics describing cost, service continuity, lifecycle behavior, or risk belong to the economic and reliability domain. No metric simultaneously satisfies the semantic, operational, and dimensional requirements of two domains unless supplemented with additional modeling layers. Conversely, adding a fourth domain would introduce redundancy, not analytical clarity: there is no separate class of metrics that possesses its own independent units, conceptual basis, and methodological lineage. Thus, the three-domain structure represents a complete and minimal basis for the metric space.

To sum up, these observations justify treating the three domains as meta-metric categories, semantically coherent, mutually exclusive, and collectively exhaustive. They define the fundamental axes along which AI infrastructure performance must be understood, and they provide the conceptual scaffolding upon which the six-layer architectural decomposition and the unified 6×3 taxonomy.

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